

Clarifications regarding MyAwesomeDB

Updated 4 weeks ago by Kaustubh Beedkar

Please use this thread to clarify any doubts regarding the assignment.

Best,
Kaustubh

[myawesomedb](#)

 0   

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 Resolved

@55_f1 

Anonymous Atom 4 weeks ago

What is the deadline of the assignment ? Also, is this compulsory to submit this i.e. will it be graded unlike previous assignments?

 1

PAIDEMARRY SAI KAMAL 4 weeks ago

Deadline and detailed submission guidelines will be announced soon.

Yes, this assignment is a graded assignment, everyone should submit for marks.

 2 Endorsed by Instructor (Kaustubh Beedkar)

 Reply

 Resolved

@55_f2 

Anonymous Mouse 4 weeks ago

Are we supposed to code in rust?
Why rust?

 1

PAIDEMARRY SAI KAMAL 4 weeks ago

Yes, you should code in rust, as the starter code is in rust.

Rust was choosen as it doesn't have a garbage collector like Java which could be a problem in memory constraint situations as in this assignment, provided simple serialisation and deserialization interfaces for structs and has compiled time memory safety guarantees

 2 Endorsed by Instructor (Kaustubh Beedkar)

Resolved

@55_f3

**Anonymous Calc** 4 weeks ago

I am trying to run monitor getting errors like this

Input query is: Query {

```
  root: Scan(  
    ScanData {  
      table_id: "region",  
    },  
  ),  
}
```

block size is 4096

First few bytes of block 0 contains "\0\0\0\0AFRICA\0lar deposits. blithely final packages c"

Memory limit is set to 64 MB

Disk IO metrics DiskIOMetricsResult {

```
  total_reads: 1,  
  total_writes: 0,  
  total_blocks_processed: 1,  
  total_cylinders_traveled: 0,  
  total_io_time_us: 4194,  
  total_seek_time_us: 0,  
  total_rotational_latency_us: 4167,  
  total_transfer_time_us: 27,  
}
```

Error: From Monitor

Caused by:

- 0: Validation failed! for simple_scan
- 1: Program did not validate the result

content of my expected file is :

0|AFRICA|lar deposits. blithely final packages cajole. regular waters are final requests. regular accounts are according to |

1|AMERICA|hs use ironic, even requests. s|

2|ASIA|ges. thinly even pinto beans ca|

3|EUROPE|ly final courts cajole furiously final excuse|

4|MIDDLE EAST|uickly special accounts cajole carefully blithely close requests. carefully final asymptotes haggle furious|

sql is select * from region;

can any body suggest me how to fix this ?

0

PAIDEMARRY SAI KAMAL 4 weeks ago

You got the correct result for starter code,

The starter code for the database doesn't implement any operators and hence doesn't send any rows to monitor for validation.

Hence you see the error from monitor saying -> Validation failed -> Program(database) didn't validate the result (i.e. didn't do validation)

Monitor interface section in the assignment document provides information about validation.

 1

[← Reply](#)

Resolved

@55_f4



Anonymous Gear 3 weeks ago

Is the sort supposed to be a "stable" sort? I mean to ask, if say the AST is such that it has something like

```
{
  "op": "Sort"
  "on": "A",
  "asc": true,
  "child": {
    "op": "Sort",
    "on": "B",
    "asc": true
  }
}
```

[run code snippet](#)

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will the output be sorted first on A and then on B? or will it be just sorted on A?

I am asking this as I couldn't find it on the notion page. Please reference the relevant paragraph, if I may have missed this in the spec.

 0

PAIDEMARRY SAI KAMAL 3 weeks ago

Sort is not necessary to be stable. The result in this case will be sorted on A having no guarantee on order of B.

If a combined sort is required, ie Sort on A if match then on B. it must be specified in a single sort operation, using the sort specs vector in sort data. One can generate a query as such using the sort_multiple method as shown in the demo_query_printer.

 0

← Reply

Resolved

@55_f5



Anonymous Calc 3 weeks ago

are we allowed to only work with main.rs of the database directory, or we can make our directory inside database directory for modularity and simplicity ?

👍 0

PAIDEMARRY SAI KAMAL 3 weeks ago

You are free to create as many files as you wish anywhere in the whole codebase, including changing the common directory as well if required.

But try not to touch the protocols of communication between components and configs.

How do we evaluate code?

We shall take your entire codebase and generate the database program using something like

```
cargo build -r --bin database
```

Then we bring our own monitor and disk programs, whose configs files could be different as well. We want your database program to work with the protocol specified in the assignment and use the config file format as generated by the generator as in starter code.

Let me know if you need further clarifications on this.

👍 0

← Reply

Resolved

@55_f6



Anonymous Poet 3 weeks ago

When will be the submission link ready so that we can check for larger datasets.

👍 0

PAIDEMARRY SAI KAMAL 3 weeks ago

Submission link will be opened near the deadline for the correctness submission.

If you wish to check against larger datasets, you can generate TPC-H datasets of size you want from a tpch generator like [this](#).

Using the generator above, you can create a dataset using the following command

```
cargo run -r --bin tpchgen-cli -- -s 0.1 -o output
```

the above command creates at scale factor of 0.1, which was used to generate sample dataset you were given in the assignment. Use bigger scale factors for bigger datasets.

It create .tbl files, rename them to .csv and schema files are the same as given in the starter dataset.

👍 0

← Reply

Resolved

@55_f7



Anonymous Calc 3 weeks ago

While working with filter, i need clarification regarding the predicate

Each filter can have multiple predicates but how will we know the predicates are joined with AND or OR ??

Can i assume that all the predicate are joined with AND??

👍 0

PAIDEMARRY SAI KAMAL 3 weeks ago

Yes, All the predicates are joined with AND. As specified in assignment

A Filter node holds a list of predicates; a row passes only if all predicates are satisfied.

Also you can assume only like types are compared, i.e. a Int32 is compared against Int32, it can't be compared with Int64 or anything else.

👍 1

← Reply

Resolved

@55_f8



Anonymous Gear 2 weeks ago

For a database for say size n, how much disk space can we assume to be available?

👍 0

PAIDEMARRY SAI KAMAL 2 weeks ago

You can safely assume a minimum of $\max(10\text{GB}, 10*n)$.

Let us know if you find more specific bounds or a bigger limit is required, the above limit was just heuristically set to prevent disk space usage explosion.

👍 1

← Reply

Resolved

@55_f9



Anonymous Helix 2 weeks ago

I had a doubt about the expected interaction between the database binary and the monitor for large scan outputs.

I implemented Scan by reading blocks from disk, decoding rows block by block and streaming them to the monitor without storing the full table in memory. This works for small tables like nation, but for a larger table

like customer, it gets stuck after some number of blocks (around 18-22 blocks to be exact, where the total blocks for the customer table are 618).

Is the intended behavior that the monitor consumes the validation output concurrently as the database writes it, or does it wait until later? For large scans, should we stream rows directly to the monitor, or is some other approach expected?

 0

PAIDEMARRY SAI KAMAL 2 weeks ago

Monitor consumes all the output concurrently, you must stream all the row directly to the monitor. Make sure you are following the protocol right and also **flushing** the outputs to disk/monitor, otherwise it can create a deadlock.

Example: Database asks disk for block x and start reading the output from disk, but the command is stuck in database program's buffer and is never received by the disk. In such case, database is waiting on disk for output, but disk is waiting on database for commands, creating a deadlock.

You can create a private post if you need more help.

 0

PAIDEMARRY SAI KAMAL 2 weeks ago

Another case of a deadlock is when database is sending a lot of requests to the disk without read the result of the disk back or reading the results after sending all requests.

This creates a deadlock as disk is processing commands one by one. As its receiving the commands, it outputs any result back to database. But if database is not reading the output(as its sending next commands without reading back yet), here the output buffers of both disk and database are filled (as disk is stuck outputting with database not reading the results and database is stuck outputting commands to disk with disk not listening commands yet).

This is another case/kind of deadlock you might have.

 0

Anonymous Helix 2 weeks ago

The deadlock was that when the monitor calls `handle_db()` and the `validate()` function detects a mismatch between the database line and expected output line, it is supposed to exit with an error due to the bail!. However, since there is a `db_process.wait()` after the function call, the monitor keeps waiting for the DB process to end and DB process waits for the monitor to consume its output thus creating a deadlock. A correct database scan works fine but just wanted to bring this to your notice in case the intended behavior of the monitor was to terminate at a mismatch instead of getting stuck in a deadlock.

 2 **Endorsed by Instructor (PAIDEMARRY SAI KAMAL)**

Anonymous Scale 1 week ago

Can you please tell how u resolved this issue, I am flushing for every 1000 rows, still I find that the program stalls.

 0

Anonymous Helix 1 week ago

You can try adding debug prints inside the monitor file. In my case there was a mismatch in the actual and expected output. The monitor stalls instead of exiting with an error. It works fine if the output matches exactly.

 0

PAIDEMARRY SAI KAMAL 1 week ago

Hi, thanks for pointing this out, we have fixed this issue in the master branch, can you pull it let us know if its fixed.

 0

Anonymous Helix 1 week ago

Yes its working now. Thanks.

 0

 Reply

Resolved

@55_f10 

Anonymous Calc 2 weeks ago

can we have some basic demo sql query such that we can implement things like `count(*)`, adding new columns or renaming in select or we have to assume everything?

 0

 Reply

Resolved

@55_f11 

Anonymous Mouse 2 weeks ago

do we need to always explicitly set `.separator |` in sqlite export scripts as default separator is `,` ?

```
Caused by:
  0: Validation failed! for Simple Scan
  1: Expected line output
    0,ALGERIA,0," haggie. carefully final deposits detect slyly agai","
    but database returned
    0|ALGERIA|0| haggie. carefully final deposits detect slyly agai|
    error at line 1
```

 0

PAIDEMARRY SAI KAMAL 2 weeks ago

Are you using sqlite3? It could also be possible that your systems installation defaults to a coma, there is also some `.sqliterc` file which can be used to set it to default to a pipe `|` character.

 0

Anonymous Mouse 2 weeks ago

yes, I am using sqlite3.

Thanks, i created a `.sqliterc` file to set default separator.

 0

[← Reply](#)

Resolved

[@55_f12](#)

Anonymous Beaker 2 weeks ago

Will all the queries be executed in one run of cargo during evaluation. I wanted to know that if I perform an expensive operation then it will pay off if large number of queries are executed on the same run of database ? Typically how many queries will be there in the test cases.

0

PAIDEMARRY SAI KAMAL 2 weeks ago

Database executes only a single query, if you look at the monitor code, when multiple queries are provided in the config, it runs each one of them in a new instance of disk and database.

So as per the spec given, we don't support running multiple queries on the same database and disk instance. Database must exit after validating the result of given query, there is no second query.

I hope this answers the question.

0

[← Reply](#)

Resolved

[@55_f13](#)

Anonymous Gear 2 weeks ago

Can we assume that a row (even an intermediate one needed for getting the final result) in a sane implementation with some column pruning be able to fit in a single block?

1 **Endorsed by Instructor (PAIDEMARRY SAI KAMAL)**

PAIDEMARRY SAI KAMAL 2 weeks ago

Yes, you can assume that.

0

[← Reply](#)

Resolved

[@55_f14](#)

Anonymous Beaker 2 weeks ago

If I want to create an index where should I create it. Can I create it in the .bin file of the table itself ? If I do that what happens to the anon_start_block when running another instance of database process ?

0

PAIDEMARRY SAI KAMAL 2 weeks ago

Please read the assignment specs carefully before posting here, what specs or aspects of the assignment made you think you can modify the .bin files? The assignment clearly states that your program can't write to files, let alone anything below anon_start_block is read only.

Additionally even if you create your own indexes on your local bin files via modifying the generator, how are they going to be compatible with bin files we use during testing which follow the specs in the assignment and come from our own generator?

 0

Anonymous Beaker 2 weeks ago

So where should I create an index ?

 0

PAIDEMARRY SAI KAMAL 2 weeks ago

In memory or in the scratch space above the anon start block. (You will have to build the index for each query as nothing is persisted between runs).

 0

 Reply

Resolved

@55_f15



Anonymous Comp 1 week ago

I am facing an issue while running the monitor binary with the provided `main.rs` (from the `monitor` folder). When I run:

```
cargo run -r --bin monitor -- --config scratch/runtimes/tpch/monitor_config.json
```

the execution hangs after:

```
DEBUG: Done, 6733 rows emitted.
```

I noticed that if I add the following lines:

```
let _ = db_process.kill();
let _ = disk_process.kill();
```

run code snippet

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before:

```
let _ = db_process.wait();
let _ = disk_process.wait();
```

run code snippet

Visit Manage Class to disable runnable code snippets ×

then the program terminates correctly and the validation succeeds as shown below :

```
DEBUG: Done, 6733 rows emitted.
```

```
-----
Validation success! for Project over Filter Test
```

I wanted to confirm:

- Are we allowed to make this change in the `main.rs` inside the `monitor` directory?
- Or is the expectation that this issue should be fixed from the database implementation side so that the processes terminate cleanly on their own?

Any clarification would be really helpful. Thanks!

 1

PAIDEMARRY SAI KAMAL 1 week ago

We won't be using your monitor code, so changing it won't work. As stated in the assignment, your database program must exit cleanly after validating the output.

 2

 Reply

Resolved

@55_f16



Anonymous Atom 2 1 week ago

Sir, regarding the tiered performance grading, getting "Bronze" requires logically correct output within the time limit. For the higher competitive tiers (Silver/Gold/Platinum/Diamond), will you be providing any TA baseline scores or "safe thresholds" for simulated IO/execution time early on? Or is our placement strictly dependent on how the rest of the class performs?

 0

PAIDEMARRY SAI KAMAL 1 week ago

The placement is strictly dependent on class performance. The safe thresholds are per question dependent. You won't need to worry about the safe thresholds for correctness case, as they are tested on relatively smaller datasets, the safe limits in this case are to prevent non terminating/ very inefficient solutions.

For the benchmark tests, your execution might fail due to set limits, in which case you need to further optimize your solution in the coming weeks.

The first deadline of submission is only to assess correctness.

 0

 Reply

Resolved

@55_f17



Anonymous Mouse 1 week ago

When we query `get_memory_limit`, should we treat that as the **total process memory budget**, including code/binary, stack, Rust runtime/allocator overhead, parsing buffers, etc., or as the memory effectively available for our **query execution data structures**?

In particular, for memory-heavy operators like **Sort** and **Cross/Join**, are we expected to **reserve some safety margin ourselves** and use only the remaining memory for in-memory buffers / intermediate state, or can we assume the reported limit is intended to be used directly for execution logic?

 0

PAIDEMARRY SAI KAMAL 1 week ago

Its the total process memory budget, which includes all of code, stack ...etc. you are expected to keep safety margins yourselves.

 1

 Reply

Resolved

@55_f18



Anonymous Helix 7 days ago

Can we assume that each query will have an explicit output column order? Suppose we do select * from customer join order then the attributes of order could be printed before the attributes of customer depending on the internal join ordering. So we might need an explicit projection like select customer.id,order.id to make it unambiguous. Will this be ensured in the query or are we supposed to preserve the join ordering in some sense?

 1 **Endorsed by Instructor (PAIDEMARRY SAI KAMAL)**

PAIDEMARRY SAI KAMAL 7 days ago

Yes every query will have an explicit column order, added via project operator.

 0

 Reply

Resolved

@55_f19



Anonymous Beaker 5 days ago

What if an order by is not present in a test case ? Will any order be acceptable during the testing ?

 0

PAIDEMARRY SAI KAMAL 5 days ago

Yes.

You can see that in testcase branch we got a new validate_bysorting method in monitor which does that.

 0

 Reply

Resolved

@55_f20



Anonymous Mouse 2 4 days ago

is the deadline of submission 13th or 15th ? What is the policy regarding it ?

 3

Anonymous Atom 2 2 days ago

Yes sir can you please tell the exact submission deadline of 1st submission.

 0

PAIDEMARRY SAI KAMAL 6 hours ago

I hope this is resolved now